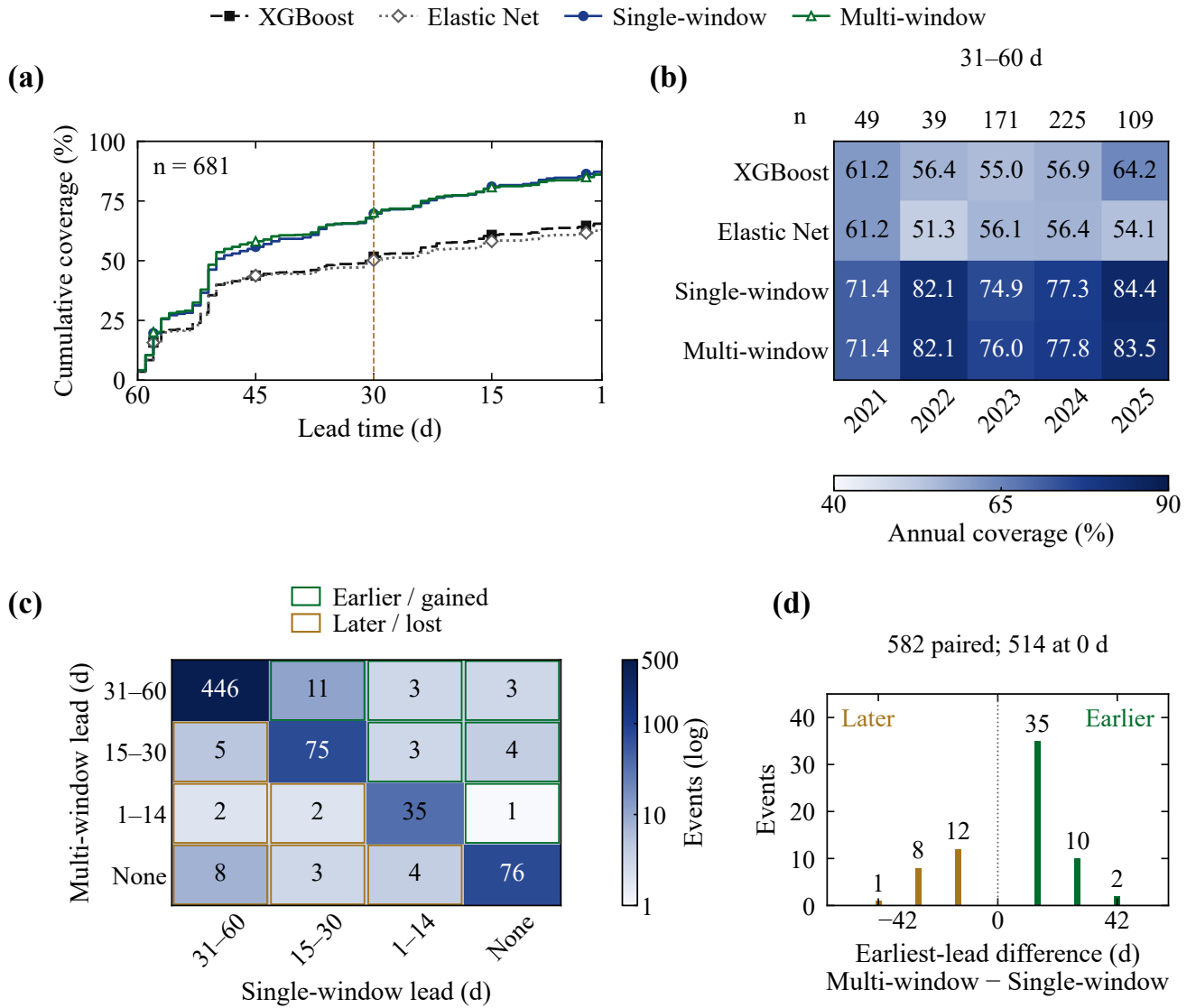

XGBoost
Elastic Net
Single-window
Multi-window
(a)
Cumulative coverage (%)
n = 681
Lead time (d)
(b)
31–60 d
n 49 39 171 225 109
XGBoost 61.2 56.4 55.0 56.9 64.2
Elastic Net 61.2 51.3 56.1 56.4 54.1
Single-window 71.4 82.1 74.9 77.3 84.4
Multi-window 71.4 82.1 76.0 77.8 83.5
2021 2022 2023 2024 2025
Annual coverage (%)
(c)
Earlier / gained
Later / lost
Multi-window lead (d)
Single-window lead (d)
Events (log)
(d)
582 paired; 514 at 0 d
Later
Earlier
Events
Earliest-lead difference (d)
Multi-window − Single-window